Investigation of The NBTI and PBTI Effects on Multiplexer Circuit Performances

M. S. S. Mohamad Saofi
Faculty of Electrical Engineering
Universiti Teknologi MARA
Shah Alam, Malaysia
safuanshafiq12345@gmail.com

H. Hussin
Faculty of Electrical Engineering
Universiti Teknologi MARA
Shah Alam, Malaysia
hanimh@uitm.edu.my

M. Muhamad
Faculty of Electrical Engineering
Universiti Teknologi MARA
Shah Alam, Malaysia
maizan@uitm.edu.my

Y. Abdul Wahab
Nanotechnology & Catalysis Research Centre, Deputy Vice Chancellor (Research & Innovation)
University of Malaya
Kuala Lumpur, Malaysia
yasminaw@um.edu.my

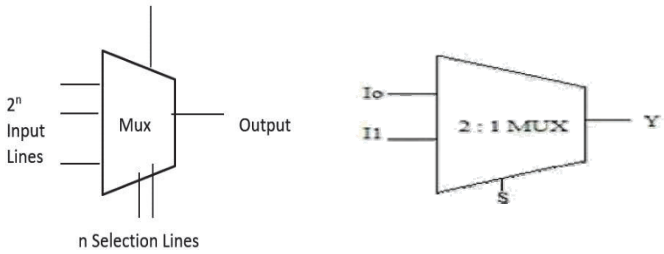
***Abstract*— This paper investigates the effects of Negative Bias Temperature Instability (NBTI) & Positive Bias Temperature Instability (PBTI) on 2-to-1 multiplexer circuit performances. The key objective of this research is to analyse the effects of NBTI and PBTI on the delay and average power of 2-to-1 multiplexer circuit. The study is conducted based on different recovery and stress simulation time condition. The performances are also studied by looking at different defect mechanisms that significantly affect the circuit's performance. The defect mechanisms that are studied in this work consists of interface trap, NIT and combination of interface oxide, NIT and oxide trap, NOT. This work used MOSRA and 32nm High Performance Model (PTM) for the reliability simulation and circuit design using Synopsis HSPICE platform. Based on the result, higher degradation can be observed if the interface trap is accounted as the defect mechanism. Simulation with combination of NBTI and PBTI shows higher degradation as compared to only NBTI or PBTI. This study further proves that NBTI shows more degradation as compared to PBTI. It is important for circuit designer to properly project the lifetime of the circuit by using appropriate defect mechanism model and considering both NBTI and PBTI effects.**

Keywords—NBTI, PBTI, Multiplexer, delay, CMOS

I. Introduction

Multiplexer is a device that allows one of several input signals selected and transmitted into a single medium, acting as both multiple-input and single-output switch. A 2- to-1 multiplexer consists of one selector, two inputs which is 10 and 11 and one output Y. To implement the multiplexer circuit, software DSCH and 32nm technology have been used to create 2-to-1 multiplexer [1]. Figure 1 shows multiplexer general circuit, the truth table and the key concept of the 2-to-1 multiplexer, where n is the number of the selector, consists 2 input signal and one output. The trials of designing integrated circuits (ICs) are concentrating on achieving high reliability and high-performance circuits. Hence, it is very crucial to understand the reliability issues and considering them when projecting the lifetime of the circuits. Current reliability issues are the NBTI and PBTI effects that significantly reduced the MOSFETs reliability hence reduce the integrated circuit's performance.

Usually, PBTI effects occur in n-channel MOSFET devices while NBTI effects usually occur in p-channel MOSFET devices. These two issues have become the key limiting factor of circuit lifetime. The performance of CMOS logic circuits can be seen and understand when NBTI and PBTI studies are conducted by looking at various operation conditions, such as the supply voltage, temperature, and node switching activity that contributes to the shift of threshold voltage hence reduce the drain current. The threshold voltage shifts depend on voltage stress on the gate oxide, temperature and the duty cycle of the stressing voltage [2].



(a) (b)

Select, S	Input		Output, Y
	I1	I0	
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

(c)

Fig. 1 (a) Multiplexer general circuit (b) 2-to-1 multiplexer concept (c) Truth table

PBTI proves to be one of the major causes of concern that requires attention. Various theories have been proposed to explain PBTI. The most common one is pre-existing traps that trapped the electrons when tunneling across interface

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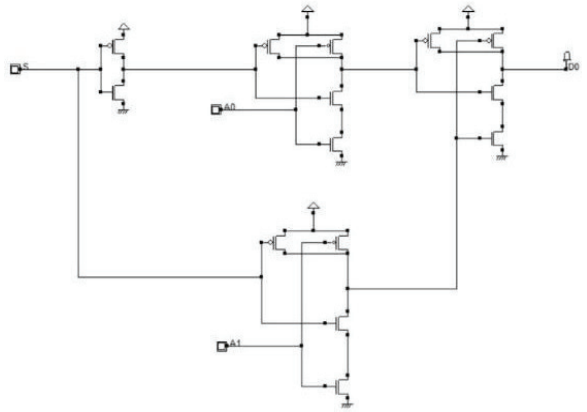
layer (IL) into high-k bulk, another one highlight that the stress induced defects become traps and trap tunneling electrons. The combination of the generation of interface state with electron trapping is proposed for devices with ultrathin gate dielectrics [3][4]. Whereas NBTI that occurs in PMOS where the dissociation of Si-H bond along the silicon-oxide interface is usually caused when the gate of the PMOS transistor is reverse biased causing the interface trap generation. These trappings can be sped up with the increment in temperature and duration of stress applied. The threshold voltage then increases caused by the traps. The degradation of circuit delay caused by enhancement of threshold resulting in failure to meet specifications after the degradation exceeds a certain level [5], [6].

Previous research shows PBTI and NBTI mechanisms occurs due to interface and oxide trapping. Referring to the R-D model, dissociation of Si-H bonds at the Si/SiO_2 interface along with the movement of H away from the interface causes the interface trap [5]. The power law of the stress time with exponent of n = 0.25 is due to the generation of the interface trap according to the R-D model [7]. Considering dynamic stressing, after forward reaction, the reverse reaction moves in motion, subsequently result in the recovery of interface traps. A "diffusion hole" is produced when the H near the Si/SiO_2 interface galvanizes rapidly. The recovery of interface trap happen upon the reverse reaction hence further result in the pushing of the diffusion towards the S/SiO_2 interface from the oxide bulk [7]. Power - law time exponent, n = 0.167 is proposed for the recovery of N_{IT} and N_{OT} effect [8]–[10] .

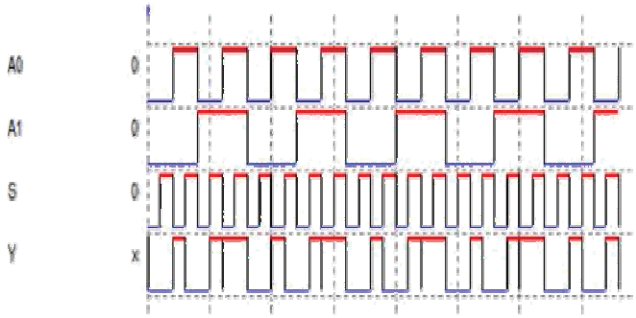
The central objective of this paper is to investigate the performance of 2-to-1 multiplexer due to PBTI and NBTI effects. The performance will be analyzed based on due to the generation of interface trap and oxide trap effects. The effects of different switching time which represent the stress and recovery timing will be investigated. This study will further probe the degradation level due to PBTI, NBTI and combination of both.

II. METHODOLOGY

In this work, 2-to-1 multiplexer is designed based on 32 nm High Performance Predictive Technology Model. Figure 2 shows the 2-to-1 CMOS multiplexer schematic circuit and the output timing diagram. For reliability simulation study, MOSFET Reliability Analysis (MOSRA) is incorporated and simulated using Synopsys HSPICE platform.



(a)



(b)

Fig. 2: (a) 2-to-1 CMOS multiplexer circuit (b) Output of timing diagram for 2-to-1 multiplexer

A. MOSRA model and Mathematical relationship

In this work, two defect mechanisms are used to properly study the continuous degradation of device characteristics. The two defect mechanisms are due to contribution of interface trap, Nit and the other one is related to the traps inside the dielectric layer, oxide trap, Not. The equation (1) below is based on interface trap contribution while equation (2) is based on oxide trap of the device. By using both equations the threshold voltage shift can be calculated. The PBTI & NBTI degradation of a circuit can also be indicated by the equation [11].

$$\Delta V_{th,IT} \approx exp\left(-\frac{Ea}{K.T}\right).\left[\frac{\varepsilon}{t_{ox}}\left(V_{gs}-V_{TH}\right)\right]^{TITCE}$$

$$.exp[TITFD.E(Vgs,Vds)]t^{NIT} \quad (1)$$

$$\Delta V_{th,OT} \approx \exp\left[-\frac{TOTFD+\frac{TOTTD}{T}}{E(Vgs,Vds)}\right]t^{NOT} \quad (2)$$

Eq. 1 and 2 show the threhold voltage shift due to interface trapped charges ($\Delta V_{t\square,IT}$) and oxide trapped charges ($\Delta V_{t\square,OT}$). The process parameters are also taking into account by fitting parameters such as TITFD (Oxide electric field dependence for interface trap inducing threshold voltage degradation), TOTFD (Oxide electric field dependent component for oxide-trap-inducing threshold voltage degradation), TOTTD (Temperature dependent component for oxide-trap-inducing threshold voltage degradation), TITTD (Temperature dependent component of interface-trap-inducing threshold voltage degradation), TITCE (Inversion charge exponent for interfacetrap-inducing threshold voltage degradation) [11].

B. Reliability simulation setup

The performance of the 2-to-1 multiplexer circuit is simulated under three types of dynamic NBTI and PBTI simulations. The temperature is fixed at 25° C. The dynamic simulation conditions are shown in Table 1 [12]. Command 1 is based on longer stress time, command 2 based on stress and recovery time alternate with similar duration of stress and recovery time, while command 3 is based on the longer recovery time. The voltage used is 1.0 V while the duration of simulation is 10 years. For verification purpose, the threshold voltage based on device degradation in inverter circuit is extracted. The verification is to ensure the time

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exponent n ~ 0.25 when interface trap accounted as defect mechanism and, n ~ 0.16 when interface trap and oxide trap considered as the defect mechanism. The circuit delay time and average power are recorded for every 1 year from fresh simulation (0 year) until aged simulation (10 years).

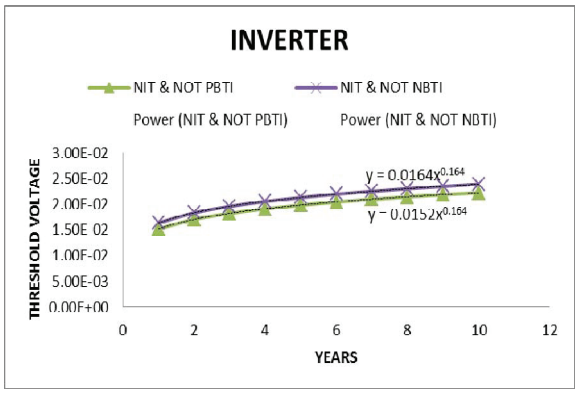
TABLE I. SIMULATION CONDITION SETUP

Command 1	.MOSRA Reltotaltime = (10*365*24*60*60)' relstep = '(10*365*24*60*60)/10' .TRAN 0.1ns 30ns
Command 2	.MOSRA Reltotaltime = '(10*365*24*60*60)' relstep = '(10*365*24*60*60)/10' +AgingStart=10ns AgingStop=100ns +AgingPeriod=1s .TRAN 0.1ns 30ns
Command 3	.MOSRA Reltotaltime = '(10*365*24*60*60)' relstep = '(10*365*24*60*60)/10' +AgingPeriod=1s AgingWidth=0.5ns .TRAN 0.1ns 30ns

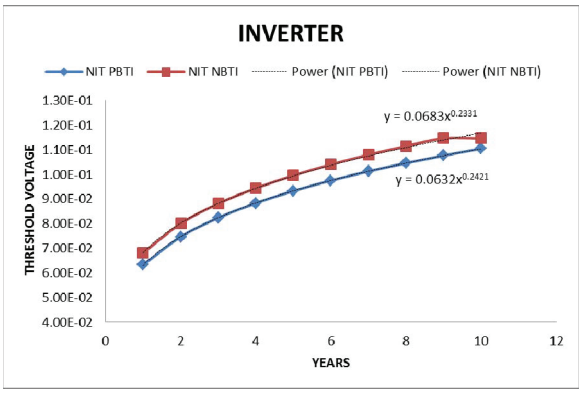
III. RESULT AND DISCUSSION

A. *Simulation verification based on threshold voltage shift of CMOS circuit*

To validate the NBTI and PBTI using MOSRA model, the time exponent, n of threshold voltage shift is used. When an inverter is simulated using NIT, the threshold voltage exhibits time exponent, n = 0.25 while combination of both NIT and NOT exhibit time exponent, n = 0.167 as seen in Figure 3. NBTI shows higher threshold voltage shift as compared to PBTI hence can be concluded that NBTI effect is more severe than PBTI effect [6].



(a)

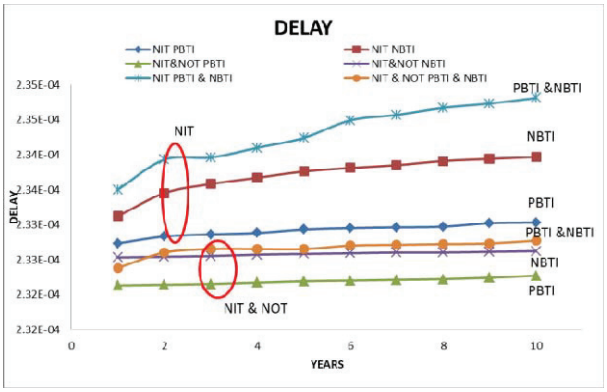


(b)

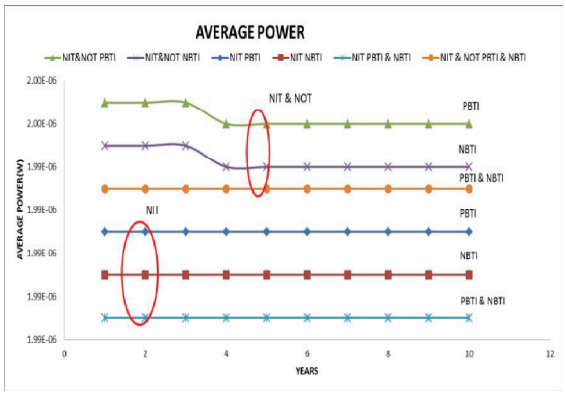
Fig. 3 (a) time exponent, n ~ 0.25 due to NIT defect mechanism (b) time exponent, n ~ 0.16 due to NIT and NOT defect mechanisms

B. *Effects of Different Defect Mechanisms*

To understand the defect mechanisms effects on the performance of the 2-to-1 multiplexer circuit, the delay and average power were studied. Figure 4 shows the effects due to different defect mechanisms on the propagation delay and average power of 2-to-1 multiplexer circuit. For this simulation, the temperature is fixed at 25°C and the simulation condition is based on command 1. Command 1 means the stress time is longer than recovery time. The results show that the delay for BTI which is consists of NBTI and PBTI is higher compared to the delay if simulation considered for NBTI or PBTI only [13]. The delay which considering combination of both defect mechanism, NIT and NOT is smaller as compared to the delay when simulated by considering only NIT as defect mechanism. This is due to oxide trap shows recovery behavior hence can reduce the PBTI and NBTI degradation [6]. As NIT is a permanent degradation component, the PBTI and NBTI effects will be increased as time increased [9]. For the average power, it is shown that the result is inverted with the delay [9]. Therefore, the effect on propagation delay is shown to be worsen when NIT is considered solely in the simulation as compared to when both NIT and NOT are considered.



(a)



(b)

Fig. 4: The effect of different defect mechanisms on multiplexer circuit performance (a) Delay (b) Average Power

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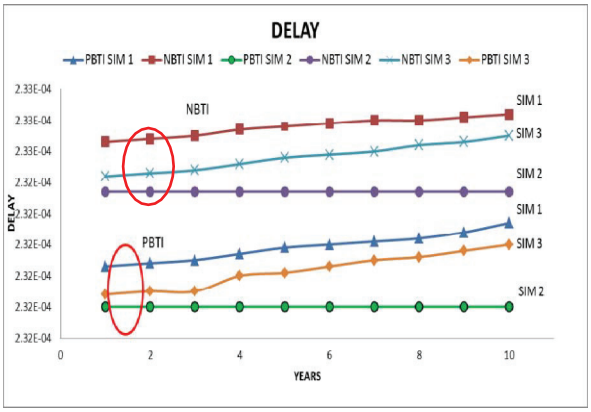
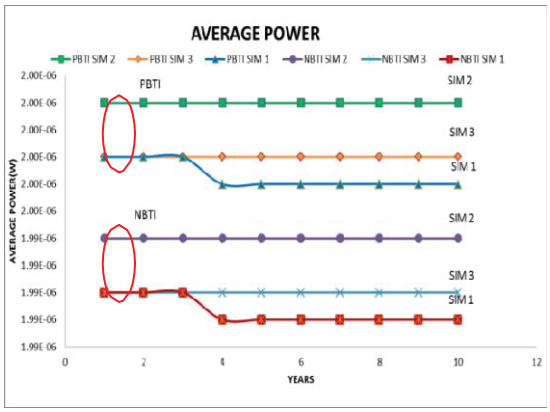
C. *Effect of Different Simulation Conditions*

Different simulation conditions study is used to understand the dynamic effect of PBTI and NBTI. The conditions as shown in Table I. To understand the dynamic effects, the defect mechanism used is fixed which based on combination of NIT and NOT. The temperature is fixed at 25°C. Figure 5 shows the delay based on command 1 and command 3 increased as the time increased while delay based on command 2 is constant. This result is due to alternate stress and recovery command that causes the degradation to be reduced and contribution by the oxide trap that exhibits recovery component of BTI. This result also shows that the effect of NBTI worse than PBTI in all simulation condition like previous result. Consequently, the decreasing of the average power is because of the increment of the delay time. The average power of the circuit such in simulation condition 2 is not affected by the short stress applied. The circuit that undergo longer stress time (command 1) show more degradation in both NBTI and PBTI effects.



(a)



(b)

Fig. 5: Effect of different simulation conditions on circuit performance (a) Delay (b) Average Power

IV. Conclusion

In conclusion, the effects of NBTI and PBTI on 2-to-1 multiplexer performance can be seen significantly. NBTI and PBTI can cause an increase of the time propagation delay. Consequently, the reduction of the average power also observed. The oxide trap shows the recovery effects subsequently reduces the NBTI and PBTI degradation. Therefore, it can be concluded that factor of the NBTI and PBTI can affect the circuit performance depends on the type of defect mechanism considered for the simulation and the duration of stress and recovery time. In addition, NBTI effects shows more degradation as compared to PBTI effects. Hence, it is important to use proper reliability model approach and understand the mechanisms to accurately project the lifetime of the integrated circuits.

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